Agentic AI Lifecycle Management

Mapping the AI Agent Lifecycle

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Course Introduction

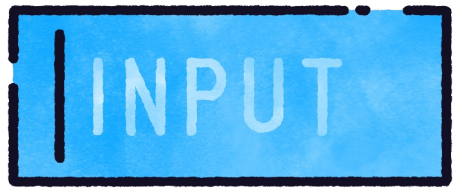
Managing the entire life cycle of AI agents

Agent's versioning and documentation for efficient maintenance

Setting policies for agent deprecation or sunsetting with plans and transition strategies

Intro to Agentic Lifecycle Thinking

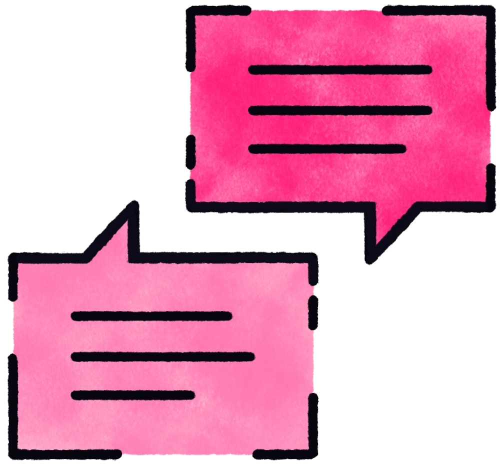
What Is Agentic AI?



Perceive inputs

Plan actions

Make decisions



Interact with
systems or users

Lifecycle Thinking in Traditional Software

Requirements	Design	Build
Test	Deploy	Maintain

Challenges without Agentic AI Lifecycle Management

Prompt inconsistencies

Model drift

Missing or changed APIs

Configuration creep

Core Lifecycle Principles

Treat agents like evolving products, not static scripts

Version everything: prompts, configs, models

Monitor for failures and silent errors

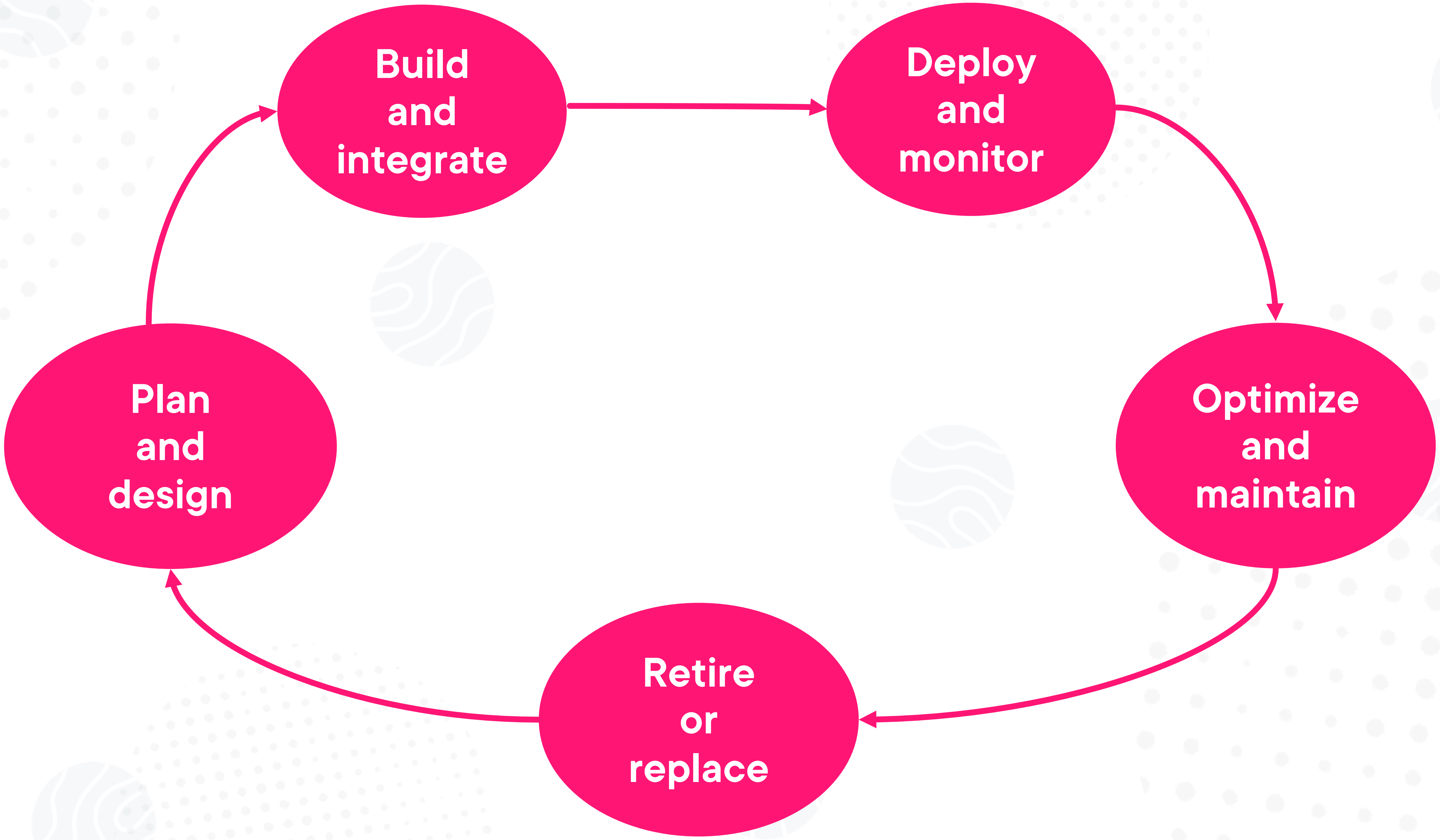
Integrate lifecycle into your CI/CD and logging stack

Lifecycle Anti-patterns

- Hardcoded prompts across the codebase
- No rollback plan for prompt/model updates
- Agents calling deprecated APIs
- No centralized error monitoring
- No ownership or documentation

Agentic Lifecycle Walkthrough

Agentic AI Lifecycle



Plan and Design

Define agent goals, roles, and success metrics

Decide will the agent use tools? Memory? APIs?

Build and Deploy

Build

Select model, craft prompts or fine-tune models, and integrate with APIs, memory, or vector search

Deploy

Launch agents in production and add observability such as logs, retries, and token tracking

Maintain and Retire

Optimize and maintain: Prompt tuning, feedback loops, address hallucinations, and retrain if needed

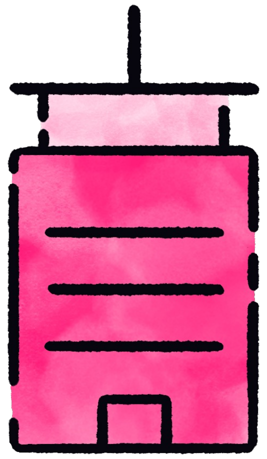
Retire or replace: Shut down or archive unused agents and hand off responsibilities

Mapping Lifecycle Phases to a Real Use Case

Globomantics deployed an AI-powered internal knowledge base agent that helps employees get quick answers to HR, IT, and operations questions by retrieving answers from internal documents and FAQs and performing actions to resolve requests and support tickets.

Globomantics’ Agent’s Responsibilities

Goal: Reduce backlog of helpdesk tickets with agentic AI

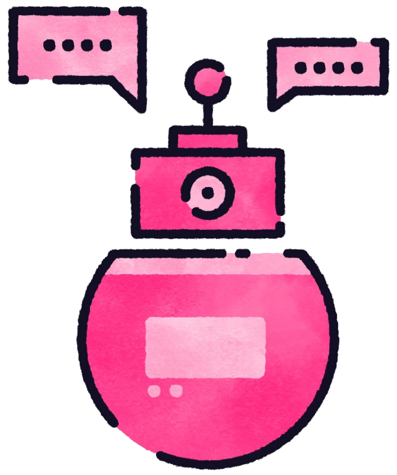


Context

Agent ingests HR and IT documentation

Receive questions

Employees ask questions



Respond

Agent responds following programmed instructions

Applying the Lifecycle

Phase	Activity Example
Plan	Identify resources, steps, and acceptance criteria
Build	Set up the tools and implementation to apply the plan
Deploy	Release the agent from the implementation source in the build phase
Maintain	Monitor failed searches, validate agent's output, and update knowledge base on a schedule
Reiterate	Apply updates, fixes, and changes while keeping in mind versioning and rollback plans
Retire/Replace	Archive documents and transfer to next-gen HR bot

Example of Possible Failures

Several key documents were updated without agent sync

Versioning conflicts with updated dependencies

If not marking resolved tickets correctly, the agent can reprocess same tickets over and over

Optimization and Automation

Configure the agent to automatically sync data and files or refresh instructions to ensure it stays up to date

Automate data and logs validation to ensure that agent responses are as expected

Fine-tune feedback and reiterate it with versioning

Set up monitoring alerts and triggers to handle failures or invalid responses

Managing Retired AI Agents

When to Retire an Agent

- Agent no longer used
- Business logic changes
- Model outdated or inefficient
- Risks outweigh value

Risks of Unmanaged Agents

- Consuming resources
- Producing confusing outputs
- Creating security holes (e.g., outdated credentials)
- Hidden dependencies

Best Practices for Retiring an Agent

- Notify dependent teams
- Archive prompt/model/version history
- Disable access tokens and credentials
- Remove from scheduled jobs
- Document in changelogs

Track Retirements

Create a retirement log or registry

Build dashboards to track agent usage

Set deactivation thresholds as needed (e.g., 30 days of inactivity)